

Chandrika Sethumadhavan

Embedded Systems Engineer | Machine Learning Engineer

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Summary

Embedded Systems Engineer with 1+ year of experience in Machine learning for sensor based and industrial applications. Proficient in Python, C, TensorFlow, and OpenCV. Published researcher with expertise in defect detection systems achieving 99.4% accuracy. Skilled in IoT data pipelines, firmware development, and deploying ML models on cloud and edge platforms.

Technical Skills

Programming: Python, C, SQL

Machine Learning: TensorFlow, TensorFlow Lite, PyTorch, YOLOv8, OpenCV, NumPy, SciPy, scikit-learn, Optuna, AutoML, TinyML

Embedded Systems: Raspberry Pi, MSP430, ESP32, PCB Design, Firmware Development, Keil uVision

IoT & Cloud: MQTT, InfluxDB, Grafana, AWS (EC2, Elastic Beanstalk), Docker, Flask

Tools: Git, GitHub, GitLab, Eagle CAD, VS Code

Experience

Master Thesis Researcher

Ongoing

IMTEK, University of Freiburg | livMatS Cluster

- Building physics-informed ML framework to predict O₂ release from ANT-EPO systems across varying temperature and concentration conditions.
- Integrating kinetic models (Henry's Law, Arrhenius) with signal processing on pressure-sensor data.
- Comparing Random Forests, XGBoost, LSTMs, and PINNs for prediction accuracy in micro-actuation contexts.

Working Student – Embedded Systems

Sep 2024 – Nov 2025

ECOSENSE Lab, IMTEK, University of Freiburg

- Designed, assembled, and tested PCBs using Eagle for environmental sensors.
- Performed firmware flashing and debugging using Keil uVision.
- Developed Python-based MQTT logging for real-time sensor data from The Things Network (TTN).
- Integrated InfluxDB Cloud and configured Grafana dashboards for live sensor monitoring.

Working Student – Image Processing & ML

Dec 2023 – Jun 2024

Fraunhofer ISE, Freiburg

- Optimized image processing pipeline for solar panel quality control using feature extraction and edge detection.
- Implemented defect detection using YOLOv8, achieving 99.4% prediction accuracy.
- Utilized OpenCV, PIL, NumPy, and SciPy for statistical analysis of mesh screen irregularities.
- Collected and analyzed data from industrial 3D Measuring Laser Microscope (LEXT).

Education

M.Sc. Embedded Systems Engineering

2023 – Present

University of Freiburg, Germany | GPA: 2.2

B.E. Electrical and Electronics Engineering

2019 – 2023

Anna University, India | GPA: 1.4

Projects

Meals or Miles: Recipe Carbon Footprint Analyzer – Black Forest Hackathon, Offenburg 2025
GitHub

- Built a recipe analysis tool that calculates meal carbon emissions and suggests sustainable ingredient alternatives using vector embeddings for multi-language ingredient matching.

- Implemented multi-objective optimization to identify key emitters (15% of total carbon) and generate nutritionally-balanced replacements, reducing carbon footprint while maintaining similar protein, carbs, and fat profiles.
- Built Flask-based backend with real-time comparison visualization. System operates with zero API calls and minimal computational overhead.

Resource-Aware AutoML for Deep Learning (Coursework)

GitHub

- Built resource-aware AutoML pipeline combining zero-cost proxies and carbon-aware techniques for efficient neural network model selection on vision datasets.
- Integrated CodeCarbon for carbon footprint tracking during hyperparameter optimization using Optuna.
- Achieved 86.26% accuracy and 0.796 F1-score on skin cancer classification using PyTorch/TorchVision.
- Evaluated across multiple datasets (flowers, emotions, fashion, skin cancer) with accuracy ranging 64.6%-97.0%.

End-to-End ML Pipeline: Academic Performance Prediction

GitHub

- Built a complete ML pipeline from data preprocessing to model training, achieving R2 scores around 0.88 with multiple ML models like Ridge, CatBoost, etc.
- Ensured robust logging and exception handling for pipeline stability.
- Developed a Flask-based web app for real-time predictions. Deployed the model on AWS using CI/CD pipelines for seamless updates. Managed code and version control with Git.

Microcontroller Programming with MSP430

GitHub

- Developed embedded systems using the MSP430 microcontroller, including LED toggling, shift registers, ADCs, interrupt handling, and timer configuration.
- Implemented Watchdog Timers (WDT) for system reset in case of deadlock, with task management through preprocessor directives for real-time operations.

Prediflex Smart Insole – Microsystems Design Lab (Coursework)

GitHub

- Virtual design and analytical modeling of an energy-harvesting microcontroller system powered by passive piezoelectric sensors that convert biomechanical energy into usable electrical power.
- The harvested energy supports both the microcontroller and wireless communication modules, conforming to international standards.
- System architecture emphasizes ultra-low-power operation, sustainable materials, and seamless integration for real-time pediatric gait analysis with predictive assessment of abnormalities.

Publications

Published in Springer (PCCDA'23) – Embedded Systems and Computer Vision

Languages

English (C2 Proficient) | German (B1 Intermediate)